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Predictive Energy Management of an AI/HPC Data Center with BESS/UPS: A Comparative Evaluation of Heuristic, Myopic Optimization, and MPC

RAONI ALDERETE¹

¹Graduate Program in Electrical Engineering, Universidade Federal de Mato Grosso do Sul (UFMS), Campo Grande, MS 79070-900, Brazil (e-mail: raoni.alderete@ufms.br)

Corresponding author: Raoni Alderete (e-mail: raoni.alderete@ufms.br).

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ABSTRACT This paper presents an applied design and comparative evaluation of a quadratic-programming model predictive control (MPC-QP) energy management system (EMS) for an artificial intelligence / high-performance computing (AI/HPC) data center with a battery energy storage system (BESS) providing UPS support. The contribution is not a new MPC theory result; it is a reproducible simulation framework and a controlled comparison protocol for critical-load energy management under AI-typical demand steps and source-limit saturation. At each sample, the EMS computes a BESS power reference that supplies the critical load, respects the main source limit, smooths source ramps, preserves state of charge (SOC), and anticipates detectable or schedulable events in the horizon. Forecast inputs are separated from the optimizer: the same plant and metrics are used to compare a reactive heuristic, a foresight-enabled heuristic (Heur+FS), a one-step myopic QP, and the event-anticipating MPC-QP. In the nominal load-variation cases, MPC-QP removes the transient unserved-power residual (0 kW), whereas heuristic and myopic QP remain limited by converter dynamics (91 kW to 103 kW). Foresight ablations, a multi-burst profile, forecast-error tests, and sensitivity analyses isolate anticipation versus multi-step optimization and delimit the physical role of BESS/UPS coordination—including that mistimed foresight can worsen the peak relative to reactive control. No hardware-in-the-loop (HIL) validation is performed; computational cost and aggregated-model limitations are stated explicitly.

INDEX TERMS Energy management system, model predictive control, BESS, UPS, data center, critical load, quadratic programming, comparative evaluation.

I. INTRODUCTION

THE rapid growth of artificial intelligence and high-performance computing workloads has increased the power density of data centers and strengthened continuity requirements for critical IT loads [10], [11], [22], [23]. Recent assessments highlight a steep rise in facility electricity demand driven by GPU clusters and generative-AI training/inference, with direct implications for supply contracts, feeder limits and on-site backup sizing [10]–[12]. Unlike relatively steady profiles, AI/HPC loads exhibit intense and recurring demand variations—training jobs, batch inference, GPU/accelerator bursts and queue shifts—that produce power steps and may saturate contractual or technical supply

limits [12]–[14]. In these everyday regimes, partial electrical shortfalls compromise critical digital services even without a complete grid outage. Source contingencies (loss/return) also matter, yet their energy outcome is strongly constrained by BESS physics: finite power and energy cannot replace the grid in prolonged outages [22]–[25]. Beyond local continuity, data centers are increasingly discussed as grid-interactive assets that can provide demand response and ancillary services [15]–[19]; the present work, however, focuses on plant-level BESS/UPS coordination for critical-load supply rather than market participation. The practical EMS design problem is therefore to coordinate the BESS/UPS so as to absorb AI-typical load dynamics while respecting source and converter

limits [20], [21].

Rule-based (heuristic) controllers remain common for supervisory simplicity and interpretability [8], [9]. Their reactive nature, however, limits performance under converter dynamics when demand steps are foreseeable in the operating horizon. One-step (myopic QP) optimization improves local consistency with costs and constraints, but still ignores future demand trajectories. Model predictive control via quadratic programming (MPC-QP) combines a dynamic model, an explicit prediction horizon and constraint handling, and is a natural candidate for storage-centric EMS design under variable load [1]–[7]. The MATLAB-oriented practical frameworks of Rossiter [1] and Wang [2] underpin the modeling, QP formulation and numerical validation workflow used here.

This work formulates a deterministic EMS-MPC via QP for an aggregated data-center load supplied by a main source and a BESS/UPS. Positioning is applied and comparative: the contribution is not a new MPC theory result, but (i) an operational EMS design for critical-load supply and BESS/UPS coordination, (ii) a reproducible simulation framework with fixed plant, parameters and metrics, and (iii) a protocol that separates forecast information from optimization structure. The operational objective is to manage BESS power while prioritizing critical-load supply under demand variation and source-limit saturation, with anticipation of detectable or schedulable events in the horizon (protection alarm, switching signal, HPC job schedule, planned load step)—not completely random outages. Grid contingencies are analyzed as a complementary study, to delimit honestly what storage can and cannot deliver; a 10 s early timing shift illustrates that mistimed foresight can be worse than reactive response.

Specific contributions are:

- 1) an aggregated plant model and a reproducible QP EMS-MPC formulation (decision vector, hard/soft constraints, weights, and SOC predictor with K_d/K_c);
- 2) a comparative protocol heuristic $\times$ Heur+FS $\times$ myopic QP $\times$ MPC-QP with/without foresight that isolates forecast versus multi-step optimization under identical plant dynamics;
- 3) evidence that, under nominal load-variation cases, MPC-QP removes the converter-dynamics transient residual in P_{unserved} ;
- 4) forecast-error, multi-burst, sensitivity (N_{prep} , τ_b) and computational-cost analyses, with explicit limitations (no HIL).

The central research question is whether the improvement comes merely from advance event information or from the combination of foresight with multi-step constrained optimization. To answer this, the protocol compares a reactive heuristic, a foresight-enabled heuristic, a one-step myopic QP, and the event-anticipating MPC-QP under identical plant dynamics and constraints—so that each baseline isolates one factor (Table 4).

The remainder of the paper is organized as follows. Section II presents the system model. Section III describes the

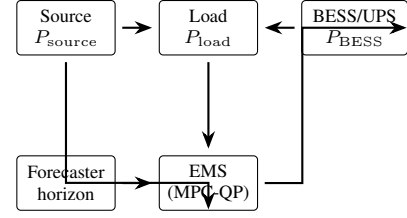


FIGURE 1. Aggregated architecture: source, critical load and BESS/UPS under the EMS-MPC, with a horizon event/load forecaster.

TABLE 1. Sign convention

Symbol	Meaning
$P_{\text{BESS}} > 0 / P_d \geq 0$	Discharge (power to the load)
$P_{\text{BESS}} < 0 / P_c \geq 0$	Charge (power from the source)
$P_{\text{source}} > 0$	Power delivered by the source
$P_{\text{unserved}} > 0$	Unserviced critical-load power
$g = 0$	Source unavailable $\Rightarrow P_{\text{source}} = 0$

controllers. Section IV defines scenarios and metrics. Section V discusses results. Section VI concludes the paper.

II. SYSTEM MODEL

A. ARCHITECTURE

An aggregated critical load $P_{\text{load}}(k)$ is supplied by a main source with limit $P_{\text{source}}^{\text{max}}(k)$ and binary availability $g(k) \in \{0, 1\}$, and by a BESS/UPS with power $P_{\text{BESS}}(k)$. Fig. 1 summarizes the information flow: the Source–Load–BESS/UPS plant is measured by the EMS, which queries a forecaster (load, limit and availability over the horizon) and issues the reference $u(k)$ to the converter. Table 1 summarizes the sign convention. Power balance at time k uses the updated converter power $P_{\text{BESS}}(k+1)$ after (2):

$$P_{\text{source}}(k) + P_{\text{BESS}}(k+1) + P_{\text{unserved}}(k) = P_{\text{load}}(k), \quad (1)$$

where $P_{\text{unserved}} \geq 0$ is computed after power, SOC and source-limit saturations.

B. BESS/CONVERTER DYNAMICS

Actual BESS power follows an aggregated first-order model:

$$P_{\text{BESS}}(k+1) = a_b P_{\text{BESS}}(k) + b_b u(k), \quad (2)$$

with $u(k) = P_{\text{BESS}}^{\text{ref}}(k)$,

$$a_b = e^{-T_s/\tau_b}, \quad b_b = 1 - a_b, \quad (3)$$

where T_s is the sampling period and τ_b is the aggregated converter time constant. After the dynamic update, P_{BESS} is saturated in $[P_{\text{BESS}}^{\text{min}}, P_{\text{BESS}}^{\text{max}}]$ and protected by SOC limits.

C. SOC UPDATE

1) Simulated plant

SOC on the plant is updated from the net energy exchanged over T_s , converted to a percentage of E_{bat} . Distinct discharge

(η_d) and charge (η_c) efficiencies are used (both 0.95 in the case study). With $P_b^+ \triangleq P_{\text{BESS}}(k+1)$,

$$\Delta \text{SOC}(k) = \begin{cases} \frac{P_b^+ T_s \cdot 100}{3600 E_{\text{bat}} \eta_d}, & P_b^+ \geq 0, \\ \frac{P_b^+ T_s \eta_c \cdot 100}{3600 E_{\text{bat}}}, & P_b^+ < 0. \end{cases} \quad (4)$$

and

$$\text{SOC}(k+1) = \text{sat}(\text{SOC}(k) - \Delta \text{SOC}(k), \text{SOC}_{\min}, \text{SOC}_{\max}). \quad (5)$$

2) Internal MPC-QP / myopic QP predictor

To keep the QP convex and reduce the plant–predictor mismatch on charging, distinct gains

$$K_d = \frac{T_s}{3600 E_{\text{bat}} \eta_d} \cdot 100, \quad K_c = \frac{T_s \eta_c}{3600 E_{\text{bat}}} \cdot 100 \quad (6)$$

are applied by successive linearization. Linearization is performed once per control step (at each sampling instant k), when the QP is assembled, and is recomputed every sample from the measured/plant power $P_{\text{BESS}}(k)$ and the free-response trajectory $P_{\text{BESS}}^{\text{free}}$ over the horizon—the open-loop response under the previous reference held constant, with no dependence on the previous QP solution for the sign choice. At each predicted step i , the sign of $P_{\text{BESS},i}^{\text{free}}$ selects the K_d (discharge) or K_c (charge) branch; with these gains fixed, the SOC map remains linear in the decision variables, so the QP remains convex. This removes the systematic $\approx 1/\eta^2$ charging error of a single-gain predictor without increasing the QP dimension. The plant continues to apply (4) with saturations.

D. CASE-STUDY PARAMETERS

Nominal parameters are summarized in Table 2. QP weights (Table 3) impose an approximate priority hierarchy via normalized magnitudes: load supply and event support dominate source tracking, SOC, ramps and BESS usage.

III. CONTROL STRATEGIES

Each controller answers a distinct design question (Table 4): the heuristic measures the performance of a simple reactive operating rule; Heur+FS measures how much is gained from future information alone, without multi-step optimization; the myopic QP measures how much is gained from constrained optimization without a future horizon; MPC-QP measures the combined effect of foresight, a dynamic model and multi-step optimization. The comparison is therefore structured: it isolates forecast versus optimization under the same plant.

A. HEURISTIC CONTROLLER

The heuristic implements supervisory rules:

- if $g(k) = 0$, then $u(k) = \min(P_{\text{load}}(k), P_{\text{BESS}}^{\text{max}})$;
- if $P_{\text{load}}(k) > P_{\text{source}}^{\text{max}}(k)$, the BESS covers the deficit;

- if source headroom exists and $\text{SOC} < \text{SOC}_{\text{ref}}$, the BESS recharges up to a charge-power limit;
- otherwise, $u(k) = 0$.

This baseline is interpretable and strictly reactive (no forecast).

B. HEURISTIC WITH FORESIGHT (HEUR+FS)

To isolate the value of prediction, a foresight-enabled heuristic (Heur+FS) receives the same future information as the MPC-QP (load, limit and g over N_{prep}). Before a detected event it applies a proximity-proportional preparation ramp; at contingency onset it uses the same reactive rules. It does not solve a QP, thus comparing forecast + rules versus forecast + multi-step optimization.

C. MYOPIC QP (ONE-STEP OPTIMIZATION)

At each k , the myopic QP solves a QP for the current instant only:

$$z = [u, P_{\text{grid}}, s_{\text{load}}, s_{\text{source}}, s_{\text{rech}}]^T. \quad (7)$$

One-step BESS power is predicted by (2) and the balance

$$P_{\text{grid}} + P_{\text{BESS}}(k+1) + s_{\text{load}} = P_{\text{load}}(k) \quad (8)$$

is imposed, together with power/SOC limits (gains K_d/K_c), source tracking, ramps and u rate limits. There is no horizon and no event preparation.

D. EVENT-ANTICIPATING MPC-QP

MPC-QP optimizes Δu over N_c with prediction over N_p . The decision vector is

$$z = [\Delta U; P_{\text{grid}}; S_{\text{load}}; S_{\text{source}}; S_{\text{rech}}; S_{\text{event}}], \quad (9)$$

with dimension $N_c + 5N_p$ (nominally $15 + 5 \cdot 60 = 315$). BESS references reconstruct as

$$u_{k+i|k} = u(k-1) + \sum_{j=0}^{\min(i, N_c-1)} \Delta u_{k+j|k}. \quad (10)$$

P_{BESS} follows (2); predicted SOC uses K_d/K_c by successive linearization. Predictive balance is

$$P_{\text{grid},i} + P_{\text{BESS},i} + S_{\text{load},i} = P_{\text{load},i}, \quad (11)$$

with $g_i = 0 \Rightarrow P_{\text{grid},i} = 0$.

1) Event detection and minimum support

Define $\delta_i = \max(P_{\text{load},i} - P_{\text{source},i}^{\text{max}}, 0)$ and

$$P_{\text{BESS},i}^{\text{min,evt}} = \begin{cases} \min(P_{\text{load},i}, P_{\text{BESS}}^{\text{max}}), & g_i = 0, \\ \min(\delta_i, P_{\text{BESS}}^{\text{max}}), & \delta_i > 0, \\ 0, & \text{o.w.} \end{cases} \quad (12)$$

A preparation ramp over N_{prep} precedes the first event; this window only shapes pre-actuation/tracking, while primary foresight comes from the horizon N_p and the event constraint (13). The soft constraint is

$$P_{\text{BESS},i} + S_{\text{event},i} \geq P_{\text{BESS},i}^{\text{min,evt}}. \quad (13)$$

TABLE 2. Nominal case-study parameters

Symbol / item	Description	Value
T_s	Sampling period	1 s
T_f	Short simulation horizon	1200 s
τ_b	Converter time constant	2 s
E_{bat}	BESS energy capacity	500 kW h
$P_{\text{BESS}}^{\text{max}}$	Maximum discharge	400 kW
$P_{\text{BESS}}^{\text{min}}$	Maximum charge (negative)	-250 kW
$P_{\text{source}}^{\text{max,nom}}$	Nominal source limit	800 kW
$R_{\text{source}}^{\text{max}}$	Desired source ramp	100 kW s ⁻¹
Typical P_{load}	Base / medium / high	600 / 750 / 950 kW
$\text{SOC}_{\text{min}}/\text{SOC}_{\text{ref}}/\text{SOC}_{\text{max}}$	Limits and reference	20 / 60 / 90 %
η_d, η_c	Charge/discharge efficiencies	0.95
N_p, N_c	Prediction/control horizons	60 / 15
N_{prep}	Event-preparation window	30 steps
Δu_{max}	u rate limit (normal/event)	120 / 400 kW

TABLE 3. Nominal QP weights (terms normalized by P_{base} , R_{base} or 100 %)

Weight	Term	Value
w_L	Load slack S_{load}	10^9
w_E	Event slack S_{event}	10^9
w_G	Source slack S_{source}	10^8
w_C	Recharge slack S_{rech}	10^6
w_P	Source tracking	$3 \cdot 10^3$
w_S / w_T	SOC horizon / terminal	$1.2 \cdot 10^3 / 1.2 \cdot 10^4$
w_R	Source ramp	300
w_B	BESS usage	20
$w_{\Delta u}$	Δu variation	50

TABLE 4. Role of each baseline in the comparison (FS: foresight; Hor.: horizon)

Controller	FS?	QP?	Hor.?	Role
Heuristic	No	No	No	reactive
Heur+FS	Yes	No	Partial	forecast effect
Myopic QP	No	Yes	No	local QP
MPC-QP	Yes	Yes	Yes	proposed

2) Cost function

With normalized variables,

$$\begin{aligned}
J = & w_L \|\tilde{S}_{\text{load}}\|^2 + w_G \|\tilde{S}_{\text{source}}\|^2 + w_E \|\tilde{S}_{\text{event}}\|^2 \\
& + w_P \|\tilde{P}_{\text{grid}} - \tilde{P}_{\text{ref}}\|^2 \\
& + w_S \|\tilde{\text{SOC}} - \tilde{\text{SOC}}_{\text{ref}}\|^2 \\
& + w_T (\tilde{\text{SOC}}_{N_p} - \tilde{\text{SOC}}_{\text{ref}})^2 \\
& + w_R \|\Delta \tilde{P}_{\text{grid}}\|^2 + w_B \|\tilde{P}_{\text{BESS}}\|^2 \\
& + w_{\Delta u} \|\Delta U\|^2 + w_C \|\tilde{S}_{\text{rech}}\|^2.
\end{aligned} \tag{14}$$

The terminal term is an SOC cost only; it is not a terminal set nor a formal stability certificate.

3) Implementation

We use $N_p = 60$, $N_c = 15$, $N_{\text{prep}} = 30$, $T_s = 1$ s. The QP is solved with MATLAB quadprog (interior-point-convex), with constraint preallocation and warm-start. Solver failures trigger a heuristic fallback classified by `exitflag`.

TABLE 5. Multi-burst profile (intervals and P_{load} levels; base 600 kW)

Interval [s]	Level [kW]	Type
[120, 180)	750	medium step
[220, 260)	950	short peak
[260, 320)	700	partial recovery
[340, 400)	950	nearby peak
[450, 600)	750	sustained step
[700, 780)	950	final peak
[800, 1000)	$600 + 40 \sin(\cdot)$	light oscillation

IV. SCENARIOS, METRICS, AND PROTOCOL

Short scenarios ($T_f = 1200$ s) are organized into three groups with distinct success criteria.

A. GROUP A — NOMINAL AI/HPC OPERATION

Target: eliminate transient P_{unserved} under demand variation (no grid outage).

- 1) **Load bands:** P_{load} switches among 600, 750, 950, 700 and 600 kW.
- 2) **Source limit:** load rises from 750 to 950 kW with $P_{\text{source}}^{\text{max}} = 800$ kW. Without BESS, the structural deficit would be $950 - 800 = 150$ kW (analytical no-EMS/ $u = 0$ reference).
- 3) **Multi-burst:** synthetic AI/HPC burst profile with the grid available ($g(k) = 1$): medium steps, nearby peaks and light oscillation (Table 5). It does not require a grid outage to evidence BESS pre-positioning before load bursts.

Success criteria (Group A): $P_{\text{unserved}}^{\text{max}} = 0$ kW, $E_{\text{unserved}} \approx 0$, no structural source shortfall beyond what the BESS covers, acceptable SOC.

B. GROUP B — SHORT CONTINGENCIES

Complementary target: approach the BESS physical floor (zero P_{unserved} is infeasible when $P_{\text{load}} > P_{\text{BESS}}^{\text{max}}$).

- 1) **Source loss:** constant 750 kW; $g(k) = 0$ on $[500, 650)$ s.
- 2) **Source return:** outage on $[400, 650)$ s with later restoration.

Success criteria (Group B): $P_{\text{unserved}}^{\text{max}}$ near the physical floor; E_{unserved} dominated by BESS power limit; SOC above SOC_{min} ; low fallback.

C. GROUP C — ENERGY BOUND

Prolonged critical outage (≈ 2 h): demonstrate that finite energy dominates the outcome; avoid overclaiming continuity.

The nominal case assumes known horizon profiles (perfect foresight in the baseline). Additional tests inject magnitude errors ($\pm 10\%$, $\pm 20\%$) and timing shifts (± 10 s) into the forecast seen by MPC.

D. ADDITIONAL STUDIES

- Foresight ON/OFF and foresight-enabled heuristic (Heur+FS).
- Sensitivity to N_{prep} and τ_b .
- Forecast-error sweeps.
- Prolonged critical outage (Group C) as an energy bound.

E. METRICS

We report $P_{\text{unserved}}^{\text{max}}$, unserved energy E_{unserved} , duration T_{unserved} , SOC (final/min/at event), throughput and equivalent full cycles N_{EFC} , maximum source ramp, fallback percentage and QP timings (mean and max per scenario). For contingencies we also use distance to the physical floor $P_{\text{unserved}}^{\text{max}} - (P_{\text{load}} - P_{\text{BESS}}^{\text{max}})$. For load variation we report transient reduction relative to the heuristic $(P_{\text{unserved,heur}}^{\text{max}} - P_{\text{unserved}}^{\text{max}})/P_{\text{unserved,heur}}^{\text{max}}$. Under outages, the maximum source ramp includes the exogenous drop induced by $g(k)$ and must not be read as a voluntary EMS operating ramp.

V. RESULTS AND DISCUSSION

The results are interpreted under two different objectives. In load-variation scenarios (Group A), the target is to eliminate transient unserved power. In contingency scenarios (Group B), the target is not zero—infeasible when $P_{\text{load}} > P_{\text{BESS}}^{\text{max}}$ —but to approach the physical BESS floor, used here as a performance benchmark.

A. MAIN COMPARISON

Table 6 summarizes $P_{\text{unserved}}^{\text{max}}$ across all short scenarios and shows these two regimes: in load-variation regimes, the performance target is zero unserved power, and MPC-QP reaches it; in source-contingency regimes, the performance target is not zero, but the physical BESS floor, which MPC-QP reaches when foresight is adequate. In the nominal load-bands and source-limit cases, MPC-QP removes the transient residual (0 kW), whereas heuristic and myopic QP exhibit peaks of about 91 kW to 103 kW linked to (2). Relative to the heuristic, transient reduction is 100 % for MPC-QP and ≈ 94.9 % for Heur+FS (load bands / source limit). Heur+FS improves the step transient when the demand change is anticipable, but without multi-step optimization it typically does not match MPC-QP. This gain is the most relevant for AI data centers, because demand steps and contractual-limit saturation are recurring—not exceptional—operating events.

TABLE 6. $P_{\text{unserved}}^{\text{max}}$ [kW] in short scenarios (after SOC-predictor correction)

Scenario	Heur.	Heur+FS	Myopic QP	MPC-QP
Load bands	90.98	4.67	102.78	0.00
Source limit	90.98	4.67	102.78	0.00
Multi-burst	90.98	60.65	105.52	0.00
Source loss	592.61	362.47	592.61	350.08
Source return	592.61	362.47	592.61	350.05

TABLE 7. Complementary metrics: load bands and source loss

Scenario	Control	$P_{\text{unserved}}^{\text{max}}$ [kW]	E_{unserved} [kWh]	SOC_f [%]	FB [%]
Load bands	Heuristic	90.98	0.064	59.51	0
Load bands	Heur+FS	4.67	0.003	59.52	0
Load bands	Myopic QP	102.78	0.036	59.25	0
Load bands	MPC-QP	0.00	0	58.80	0
Source loss	Heuristic	592.61	14.75	57.94	0
Source loss	Heur+FS	362.47	14.59	57.54	0
Source loss	Myopic QP	592.61	14.75	57.91	0
Source loss	MPC-QP	350.08	14.58	57.31	2.33

Table 7 complements with E_{unserved} , final SOC and fallback on the two key scenarios.

Under contingencies (complementary study), heuristic and myopic QP reach ≈ 593 kW, consistent with the first step from $P_{\text{BESS}} \approx 0$ and $u = P_{\text{BESS}}^{\text{max}}$:

$$P_{\text{BESS}}^+ \approx b_b \cdot 400 \approx 157 \text{ kW} \Rightarrow P_{\text{unserved}} \approx 750 - 157 = 593 \text{ kW}. \quad (15)$$

With anticipation, MPC-QP arrives at the outage already near 400 kW, reducing $P_{\text{unserved}}^{\text{max}}$ to the physical floor

$$P_{\text{unserved}}^{\text{floor}} = P_{\text{load}} - P_{\text{BESS}}^{\text{max}} = 350 \text{ kW}. \quad (16)$$

Distance to the floor on source loss is 242.61 kW (heuristic), 12.47 kW (Heur+FS) and ≈ 0.08 kW (MPC-QP). Heur+FS is near the floor on a single contingency (362.47 vs. 350.08 kW) but fails on multi-burst (60.65 vs. 0 kW): simple foresight helps for an isolated event, whereas multi-step optimization becomes decisive when several events and SOC/ramp trade-offs interact. This floor is not a controller failure: it is the limit imposed by BESS power rating. In other words, even an optimal EMS cannot invent energy or power beyond the available storage.

Table 7 shows that MPC-QP does not win on peak alone: on load bands it drives E_{unserved} to 0 kWh (vs. 0.064 kWh for the heuristic) with final SOC 58.80 % vs. 59.51 %—spending less than one additional SOC percentage point to eliminate the transient residual. On source loss, it reduces the peak to 350.08 kW with a modest SOC cost (57.31 %) and 2.33 % fallback. Zero P_{unserved} in Group A is not obtained by oversizing the source: at the source-limit step, $P_{\text{source}}^{\text{max}}$ remains 800 kW while the load reaches 950 kW (structural deficit without BESS ≈ 150 kW); the result comes from pre-positioning P_{BESS} before the step.

Fig. 2 shows the full comparison for the load-band scenario—the primary evidence of the contribution. MPC-QP anticipates demand steps, reducing P_{unserved} transients

TABLE 8. MPC-QP under source loss: foresight ON vs OFF

Mode	$P_{\text{unserved}}^{\text{max}}$ [kW]	Unserved energy [kWh]	Final SOC [%]
Foresight ON	350.08	14.58	57.31
Foresight OFF	592.57	14.75	57.81

and coordinating P_{BESS} and SOC more effectively than strategies without a horizon.

Fig. 3 reinforces the same point in the source-limit scenario (step zoom; full panels omitted for space). At the 750 $\rightarrow$ 950 kW step, MPC-QP pre-positions the BESS and keeps $P_{\text{unserved}} = 0$; the heuristic exhibits a peak of ≈ 90.98 kW. The myopic QP, although locally optimal, is worse than the heuristic in this transient (≈ 102.78 kW vs. ≈ 90.98 kW): with Δu rate limits, ramp/SOC/ Δu penalties and no future-step visibility, it does not pre-position the BESS and tends to react more conservatively than the heuristic rule, which jumps harder on the deficit.

Fig. 4 illustrates source loss as complementary analysis (full panels and source-return figures omitted for space; return replicates the same preparation mechanism). With foresight, MPC-QP raises P_{BESS} before the event, whereas heuristic and myopic QP react only at outage onset. The value here is transient mitigation down to the physical floor—not elimination of the structural energy deficit.

B. ROLE OF FORESIGHT

To isolate the value of advance information, the source-loss scenario was repeated with the same MPC-QP while masking future events in the horizon (foresight OFF). Results are reported in Table 8.

Three observations stand out. First, without foresight the peak collapses to the reactive value (≈ 593 kW): the MPC-QP advantage is not intrinsic to QP alone, but to seeing the event with lead time compatible with τ_b and N_p . Second, unserved energy changes little (14.58 vs. 14.75 kWh): for a short outage (150 s), total energy deficit is dominated by the floor $P_{\text{load}} - P_{\text{BESS}}^{\text{max}}$ over the contingency; foresight mainly shapes the initial peak, a critical metric for service quality, protection and SLA. Third, final SOC remains close in both modes (57.31 % vs. 57.81 %), indicating that anticipation consumes a modestly larger share of useful energy before the fault in exchange for reducing the severe P_{unserved} transient to about 350 kW.

In practice, foresight may come from protection alarms, switching signals, HPC/AI load forecasts or operational schedules—that is, detectable or schedulable events. Without such a signal the controller remains valid, yet behaves as a computationally expensive reactive optimizer. The method does not claim universal robustness to completely random outages.

C. SENSITIVITY

Table 9 varies $N_{\text{prep}} \in \{0, 10, 30, 45\}$ with foresight ON. In all cases, $P_{\text{unserved}}^{\text{max}}$ stays near the floor (≈ 350 kW–

TABLE 9. Sensitivity to N_{prep} (source loss, foresight ON)

N_{prep}	$P_{\text{unserved}}^{\text{max}}$ [kW]	Unserved energy [kWh]	Mean t_{QP} [s]
0	350.27	14.58	0.071
10	350.43	14.58	0.065
30	350.08	14.58	0.060
45	350.04	14.58	0.089

TABLE 10. Sensitivity to τ_b (source loss, foresight ON)

τ_b [s]	$P_{\text{unserved}}^{\text{max}}$ [kW]	Energy [kWh]	FB [%]	Mean t_{QP} [s]
1	350.03	14.58	2.58	0.045
2	350.08	14.58	2.33	0.042
4	350.02	14.58	0.08	0.044

TABLE 11. Prolonged critical outage (≈ 2 h, 750 kW load)

Control	$P_{\text{un}}^{\text{max}}$ [kW]	E_{un} [kWh]	SOC _{min} [%]	SOC _f [%]
Heuristic	750.00	1309.95	20.00	37.41
MPC-QP	750.00	1314.78	20.00	37.40

350.4 kW) and unserved energy remains ≈ 14.58 kWh. In particular, $N_{\text{prep}} = 0$ already reaches ≈ 350.27 kW: primary foresight comes from the horizon N_p and the future event constraint (13); N_{prep} only shapes the pre-event ramp/tracking. Once the event enters N_p and the constraint is active at the outage instant, temporal coupling in the dynamic model induces pre-actuation: the optimizer raises u beforehand to respect first-order dynamics. Thus N_{prep} is not the key foresight mechanism in this case.

The converter time constant (Table 10) keeps $P_{\text{unserved}}^{\text{max}}$ at the physical floor for $\tau_b \in \{1, 2, 4\}$ s in this campaign (350.02–350.08 kW). Residual fallback remains low (up to ≈ 2.6 % for $\tau_b = 1$ s and 2 s; < 0.1 % for $\tau_b = 4$ s), without peak degradation for the slower dynamics in the tested range. In design, τ_b , N_p/N_{prep} and foresight quality should be co-tuned; values outside this range may require a longer horizon or relaxed Δu in event mode.

D. PROLONGED CRITICAL OUTAGE

Under an approximately 2 h outage at constant 750 kW, heuristic and MPC-QP were compared (Table 11). Both reach SOC_{min} = 20 %. After useful BESS depletion, $P_{\text{unserved}}^{\text{max}} = 750$ kW and unserved energy remains on the order of 1310 kWh–1315 kWh. Time with unmet power coincides with contingency duration (7200 s), because the 400 kW BESS rating is below the load.

The interpretation is energetic, not merely control-theoretic. Useful energy between SOC_{ref} = 60 % and SOC_{min} = 20 % is approximately

$$E_{\text{useful}} \approx \frac{60 - 20}{100} E_{\text{bat}} \eta_d = 190 \text{ kWh}. \quad (17)$$

Even with continuous discharge at 400 kW, this reserve covers only tens of minutes of partial support; the mini-

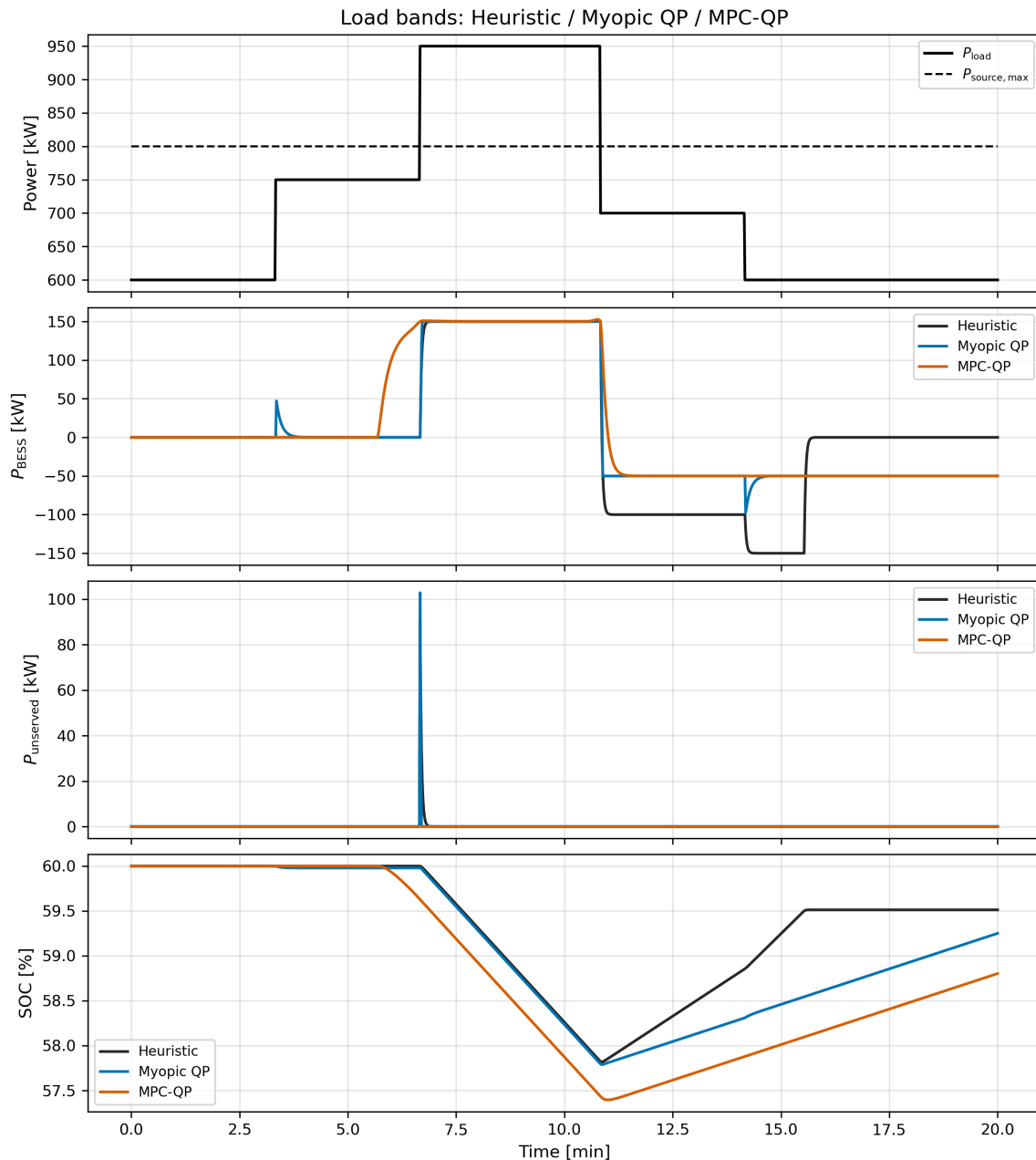


FIGURE 2. Load bands (primary evidence): heuristic, myopic QP and MPC-QP—load, source, BESS, SOC and unserved power.

mum instantaneous deficit during full BESS support remains 350 kW. Thus MPC-QP improves the initial contingency transient (when foresight is adequate), but cannot create missing energy: the controller does not invent power or energy beyond the installed BESS rating and stored charge [22], [25], [26]. In long outages, performance is dominated by nominal BESS power, available energy and backup sizing; resilience then requires a generator, a second feeder, or a larger energy reservoir rather than a more sophisticated EMS alone. The EMS role in this regime is disciplined reserve management down to SOC_{min} , avoiding premature depletion and respecting power limits—an objective both controllers

meet similarly in the prolonged-outage simulation.

E. COMPUTATIONAL ASPECTS

From the campaign CSV of the main comparison (MATLAB R2025b, Windows PC, Intel Core i7-4510U, quadprog/interior-point-convex), Table 12 summarizes mean/max MPC-QP solve times and fallback by scenario. Wall-clock simulation time is larger because it includes plant updates and logging, not only the QP. The reported solve times therefore do not demonstrate strict real-time operation at $T_s = 1$ s on this conventional PC: occasional solves approach a non-negligible fraction of the

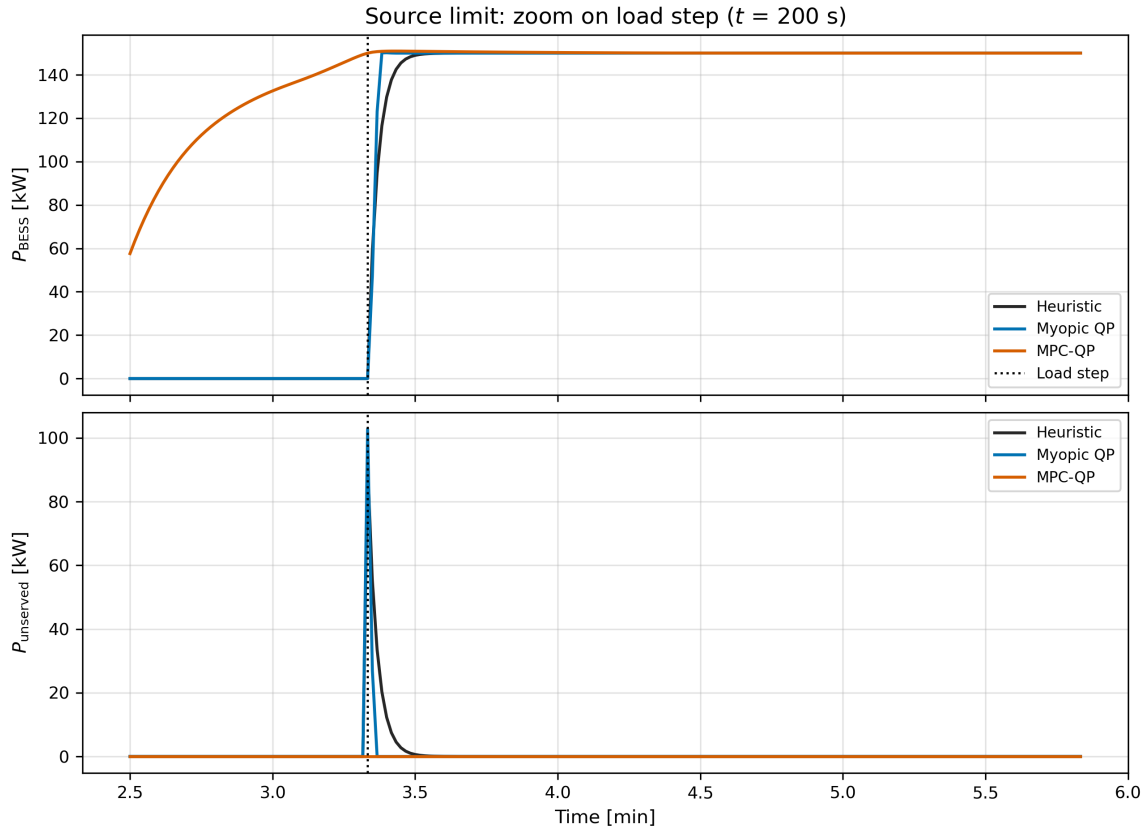


FIGURE 3. Source limit: zoom on the 750 → 950 kW step. MPC-QP removes P_{unserved} ; myopic QP (102.78 kW) exceeds the heuristic peak (90.98 kW).

TABLE 12. MPC-QP solve times and fallback (main campaign)

Scenario	Mean t_{QP} [s]	Max t_{QP} [s]	FB [%]
Load bands	0.037	0.322	0
Source limit	0.046	0.131	0
Source loss	0.044	0.192	2.33
Multi-burst	0.041	0.228	0

TABLE 13. Load-forecast magnitude error (source loss, MPC-QP)

Error [%]	$P_{\text{unserved}}^{\text{max}}$ [kW]	Energy [kWh]	FB [%]
0	350.08	14.58	2.33
−10	350.13	14.86	2.00
+10	350.17	14.58	2.83
−20	350.07	15.18	1.00
+20	350.09	14.58	2.58

sampling period, and deployment must budget computational headroom. For practical implementation we recommend (i) a supervisory EMS with larger T_s , (ii) horizon reduction (N_p/N_c), (iii) an embeddable/online QP solver with warm-start (preallocation is already used here), and (iv) future HIL validation on the target platform.

F. FORECAST ERROR AND LIMITATIONS

Under perfect foresight, MPC-QP reaches the physical floor on source loss (≈ 350.08 kW). Magnitude errors of $\pm 10\%$ and $\pm 20\%$ in the predicted load (Table 13) were tolerable: the peak remains at the floor (≈ 350 kW). Timing shift of the predicted outage (Table 14) was the most critical factor and delimits the method's scope: anticipating the fault by 10 s raises $P_{\text{unserved}}^{\text{max}}$ to ≈ 699 kW—worse than reactive (≈ 593 kW)—via premature discharge / preparation at the wrong instant. A 10 s delay degrades moderately to ≈ 357 kW. Thus the gain depends mainly on the timing

quality/lead of the foresight signal. Explicitly, the method is appropriate for detectable, programmable or schedulable events (alarms, switching signals, HPC/AI job schedules), not for completely random faults without a prior cue. This result motivates conservative event-confirmation logic before aggressive pre-positioning.

Overall limitations are consolidated as follows. The study uses an aggregated electrical plant model without detailed grid or thermal dynamics and without a detailed electrochemical degradation model; parameters were not identified experimentally on a real facility; no hardware-in-the-loop (HIL) validation was performed; and no field AI/HPC workload traces were used—forecast tests remain synthetic magnitude/timing perturbations around the nominal profiles.

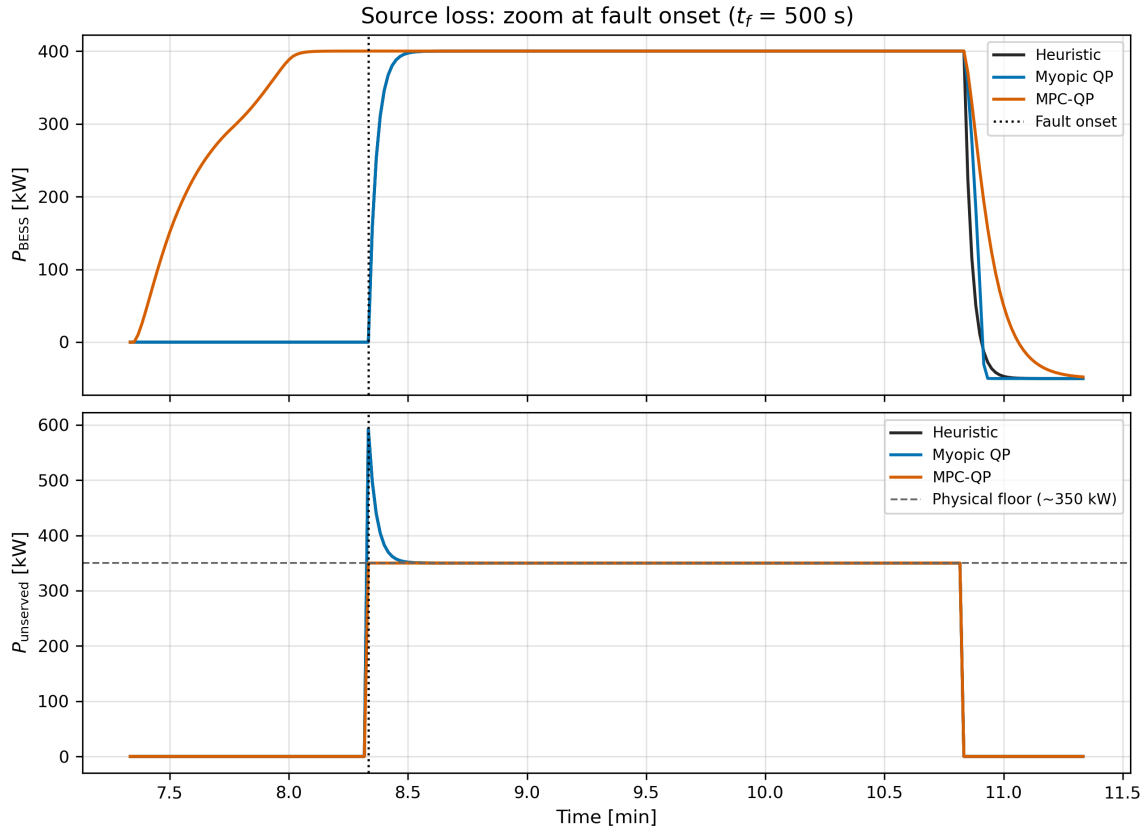


FIGURE 4. Source loss: zoom of P_{BESS} and unserved power at fault onset; MPC-QP pre-positions to the ≈ 350 kW floor.

TABLE 14. Predicted-outage timing shift (source loss, MPC-QP)

Shift [s]	$P_{\text{unserved}}^{\text{max}}$ [kW]	Energy [kWh]	FB [%]
-10	699.23	15.16	2.17
0	350.08	14.58	2.33
+10	356.86	14.59	2.17

G. CONSOLIDATED DISCUSSION

Taken together, the campaign shows that the MPC-QP gain comes from the combination of foresight and multi-step constrained optimization against converter dynamics—not from QP alone and not from foresight alone. The myopic QP does not pre-position the BESS: without a future step in the horizon it remains reactive under Δu limits and ramp/SOC penalties, and can therefore perform worse than the heuristic on the load step (102.78 vs. 90.98 kW). Heur+FS captures part of the foresight benefit on isolated anticipable events, but does not resolve multi-event trade-offs well (multi-burst peak 60.65 kW vs. 0 kW for MPC-QP). The results support four messages:

- 1) under demand steps and source-limit saturation, MPC-QP removes transient P_{unserved} in the nominal cases and outperforms heuristic and myopic QP;
- 2) the decisive mechanism is horizon- N_p anticipation with multi-step optimization; Heur+FS is only a partial

surrogate, and N_{prep} mainly shapes preparation;

- 3) under grid contingencies, MPC-QP mitigates the peak down to the BESS physical floor when foresight is adequate and well timed, but cannot overcome storage power/energy limits nor random outages without a cue;
- 4) in long outages, autonomy is dominated by energy sizing and backup supply; EMS value concentrates on reserve management and normal variable-load operation.

VI. CONCLUSION

This paper presented an applied, comparative MPC-QP EMS for AI/HPC data-center energy management with BESS/UPS, together with a reproducible simulation framework that isolates forecast versus multi-step optimization. In nominal load-variation scenarios (bands, source limit and multi-burst), MPC-QP removed the transient unserved-power peak (0 kW), while heuristic and myopic QP remained in the 91 kW to 106 kW range; Heur+FS was intermediate on anticipable steps. Under contingencies, MPC-QP reduced the initial transient to the BESS physical power floor (≈ 350.08 kW on source loss) when foresight timing was adequate; magnitude forecast errors within $\pm 20\%$ preserved that floor, whereas a 10 s early timing shift raised the peak to ≈ 699 kW (worse than reactive).

The main message is threefold. First, MPC improves pre-

dictive BESS/UPS coordination and eliminates converter-limited transients in the nominal AI/HPC load-variation cases. Second, under source contingencies the benefit depends critically on foresight timing and is bounded by the physical BESS floor. Third, the method does not replace energy sizing or backup generation (generator, second feeder, or larger storage). Future work includes HIL validation, workload-based forecasting, multiobjective BESS sizing and broader statistical analysis under uncertainty.

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RAONI ALDERETE Raoni Alderete received the B.S. degree in electrical engineering from the Universidade Federal de Mato Grosso do Sul (UFMS), Campo Grande, MS, Brazil, in 2013. He is currently pursuing the M.Sc. degree in electrical engineering at UFMS. From 2016 to 2020, he was an electrical engineer with Energisa Mato Grosso do Sul, working on distribution construction and maintenance planning. Since 2020, he has been the Founder and Technical Director of MR Energy, Campo Grande, Brazil, where he leads electrical engineering projects involving low- and medium-voltage systems, substations, photovoltaic generation, generators, and uninterruptible power supplies (UPS) for public and private critical facilities. His research interests include model predictive control, battery energy storage systems (BESS), energy management systems for artificial intelligence and high-performance computing (AI/HPC) data centers, and applied optimization for power systems.

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